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A smart system for matching resumes with job descriptions

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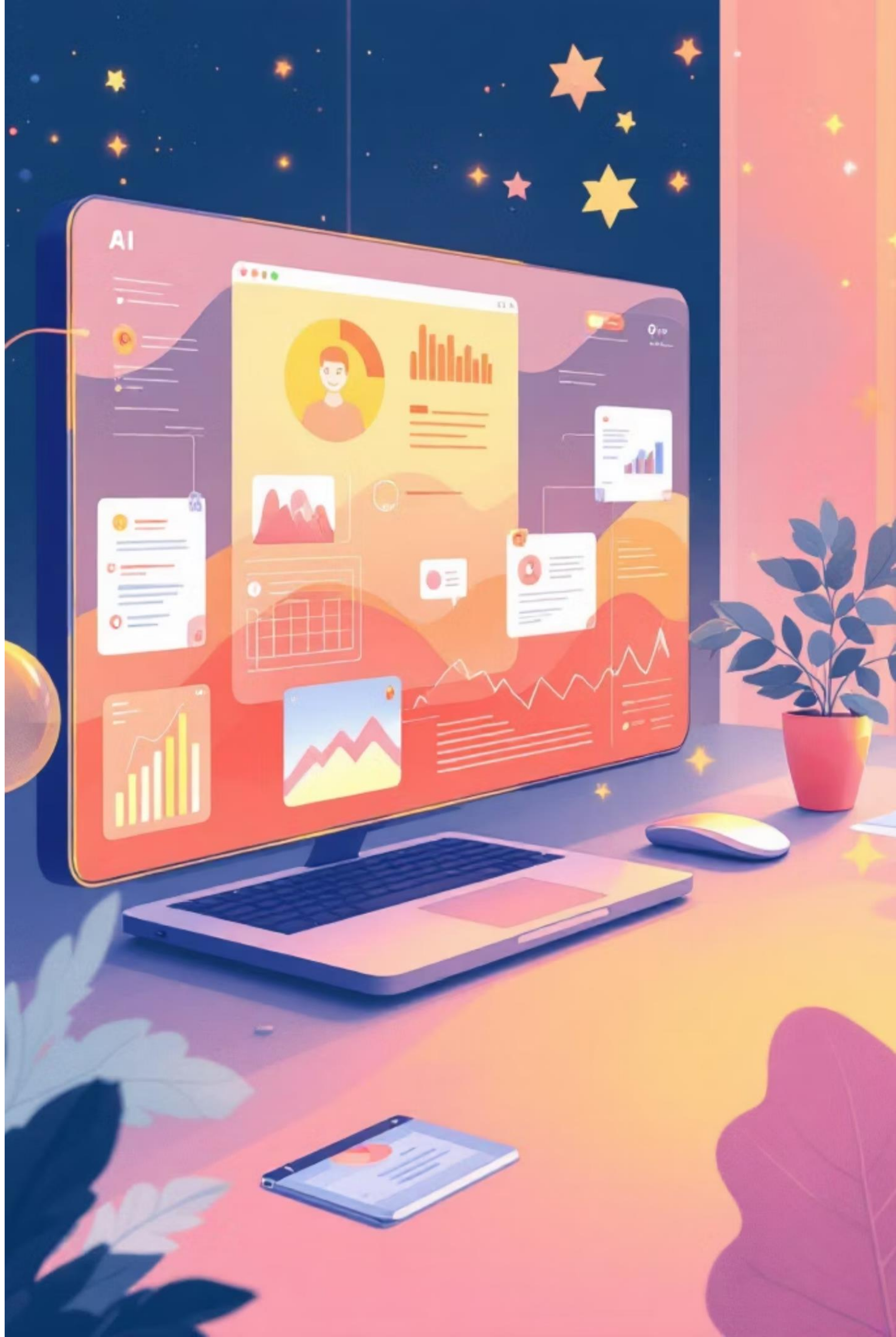


Project problem

Current challenges in the recruitment process

Semantic gap

Excluding excellent talent due to differing terminology (developer vs. programmer), where traditional systems fail to grasp the meaning.

Random data

Most resumes are disorganized, making it difficult to extract accurate information and compare applicants.

Time and effort

Thousands of resumes for each job make manual screening a slow and tedious process for HR.

Project objectives

1. Automation and speed

Speed up the sorting process and extract the best candidates with high accuracy in just a few seconds.

2. Deep understanding of meaning

Using artificial intelligence models to understand context and meaning rather than literal matching.

3. Reliance on organized data

Avoid text clutter by using a database divided into clear columns.

No.	Study Name	Input	Output	Approach	Evaluation	Results	Resolved	Unresolved
1	Automatic Resume Screening Using Sbert (2025)	سير ذاتية بصيغ غير منظمة ووصف وظيفي (JD)	درجة ملاءمة احتمالية وتوصية بالكورسات	استخدام نموذج SBERT للمقارنة الدلالية مع Random Forest.	استخدام SBERT للمقارنة العميقة (Semantic)	حققت دقة مطابقة (Accuracy) تصل إلى 92% عند استخدام SBERT مع Random Forest	تجاوز مشكلة "الكلمات المفتاحية" البسيطة للفهم العميق للسياق	التحدي في معالجة لغات غير الإنجليزية أو تنسيقات معقدة جداً
2	Resume Screening Automation with NLP Techniques (2025)	ملفات PDF و DOCX لسير ذاتية	قائمة مرشحين مرتبة مع نسبة المطابقة	استخدام Cosine Similarity لتقييم التشابه بين المتجهات النصية	دقة اختيار المرشحين بناءً على المهارات والخبرات	سجلت نسبة دقة (Precision) بلغت 89% في ترتيب المرشحين الأنسب للوظيفة.	أتمتة عملية استخراج البيانات لتقليل التحيز البشري	الاعتماد على قوائم مهارات محددة مسبقاً قد يحد من الشمولية
3	Novel Approach for Detecting Resume Columns (2025)	ملفات سير ذاتية بتنسيق الأعمدة (Columns)	هيكلية نصية منظمة مستخرجة من الـ PDF	خوارزميات رؤية حاسوبية (Computer Vision) لتحديد الأعمدة	دقة ترتيب النص المستخرج (Extraction Accuracy)	دقة عالية في ترتيب النص المستخرج من الـ PDF.	تجاوز أخطاء استخراج النصوص من ملفات PDF المعقدة	لا تزال تواجه صعوبة مع الصور والرسومات التوضيحية
4	Resume Classification System using NLP & ML (2024)	مجموعة بيانات (Dataset) تحتوي على 962 سيرة ذاتية مصنفة	تصنيف السيرة الذاتية ضمن 25 فئة وظيفية مختلفة	مقارنة بين 9 نماذج ML (مثل SVM و KNN) باستخدام TF-IDF.	معايير الدقة (Accuracy)، Precision ، و F1-Score	SVM تفوق نموذج بدقة تجاوزت 96%	حل مشكلة تصنيف البيانات الضخمة بدقة عالية جداً	مشكلة البيانات غير المتوازنة (Imbalanced Data) لبعض التخصصات
5	Enhanced Resume-Job Matching via Graph-based Networks (2024)	بيانات المهارات والخبرات كـ "عقد" (Nodes)	تمثيل بياني للعلاقة بين المهارات	Graph Neural Networks (GNN) لربط المهارات المترادفة	مدى دقة الربط بين المهارات المتباعدة	تحسين المطابقة بنسبة 15% عن الطرق التقليدية	فهم أن مهارة "التنقيب عن البيانات" مرتبطة بـ "الذكاء الاصطناعي"	تحتاج لقوة معالجة عالية وبناء Graph ضخم
6	Application of LLM Agents in Recruitment (2024)	نصوص السير الذاتية والوصف الوظيفي	تقارير تحليلية آلية وقرارات توظيف	LLM Agents (مثل GPT-4) لمحاكاة خبير التوظيف	LLM Agents (مثل GPT-4) لمحاكاة خبير التوظيف	تقليل زمن المراجعة الأولية بنسبة 80%	القدرة على شرح "لماذا" تم اختيار هذا المرشح	تكلفة استخدام الـ API لنماذج LLM الضخمة

Tools and techniques used

Data Analysis

Pandas, Matplotlib, Seaborn

Development Environment

Google Colab or Jupiter Notebook for experimentation and implementation

Programming Language

Python for ease of use and data processing capabilities

Artificial intelligence models

XGBoost, Random Forest, ,Sentence BERT

Voting Classifier , Gradient Boosting

Cleaning text

re library for text processing

Data set

(Resume Dataset by Saugata Roy Arghya)

Source

A global, open-source database from the Kaggle platform, specializing in applicant profile data across various professional sectors.

Size

More than 9,000 resumes (records), organized and structured into 35 columns (features) to facilitate data analysis.

Processing: Cleaning and converting data into binary categories (0 for rejection and 1 for acceptance) based on a threshold of 0.6.

Balancing: Using Scale Pos Weight techniques to address data imbalances (Class Imbalance).

Workflow pipeline: From raw data to decision

2

Cleaning data

Converting unstructured data into a standardized format and removing noise

1

Data exploration

Initial data analysis using EDA to understand distributions and underlying patterns

4

Models

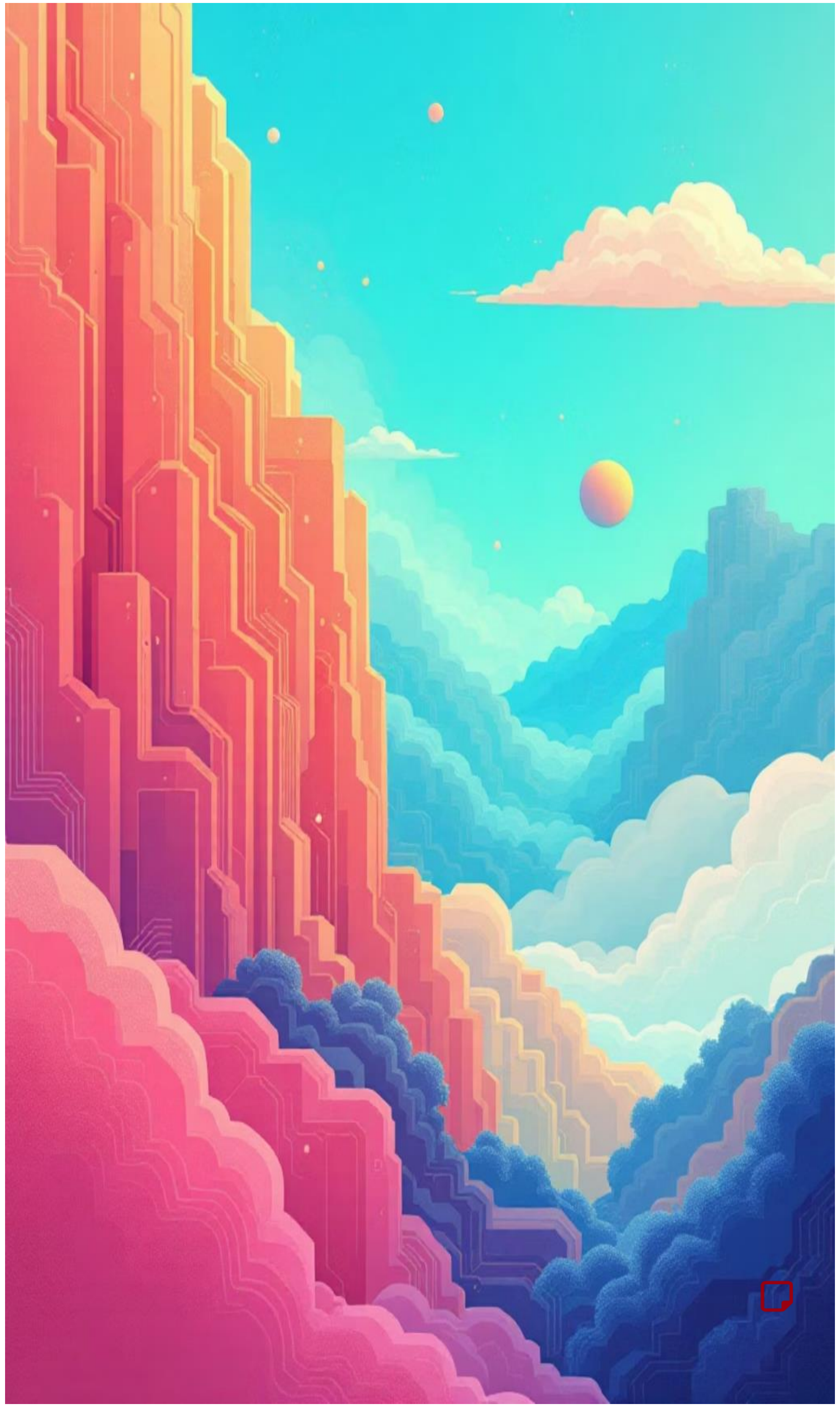
Combining multiple models to obtain accurate and reliable predictions

3

Feature engineering

Extracting multidimensional features from experience, academics, and skills

The process begins with CSV resume files and goes through a series of intelligent transformations, ultimately leading to data-driven hiring decisions. Each stage builds upon the previous one to ensure the quality and accuracy of the final results.



Why SBERT & XGBoost

Sbert →

Contextual Understanding: Captures the deep meaning of sentences rather than just matching keywords.

Computational Efficiency: Fast at generating sentence embeddings, making it ideal for real-time resume screening.

Synonym Recognition: Automatically recognizes that "Architectural Designer" and "Building Consultant" refer to similar roles.

Xgboost →

Robust on Structured Data: Superior performance in handling tabular features like "Years of Experience" and "Semantic Scores.“

Imbalance Handling: Effectively manages the scarcity of "Suitable" candidates using weighted logic like scale_pos_weight.

Regularization & Accuracy: Built-in mechanisms to prevent "Overfitting,“ ensuring the model generalizes well to new, unseen CVs.

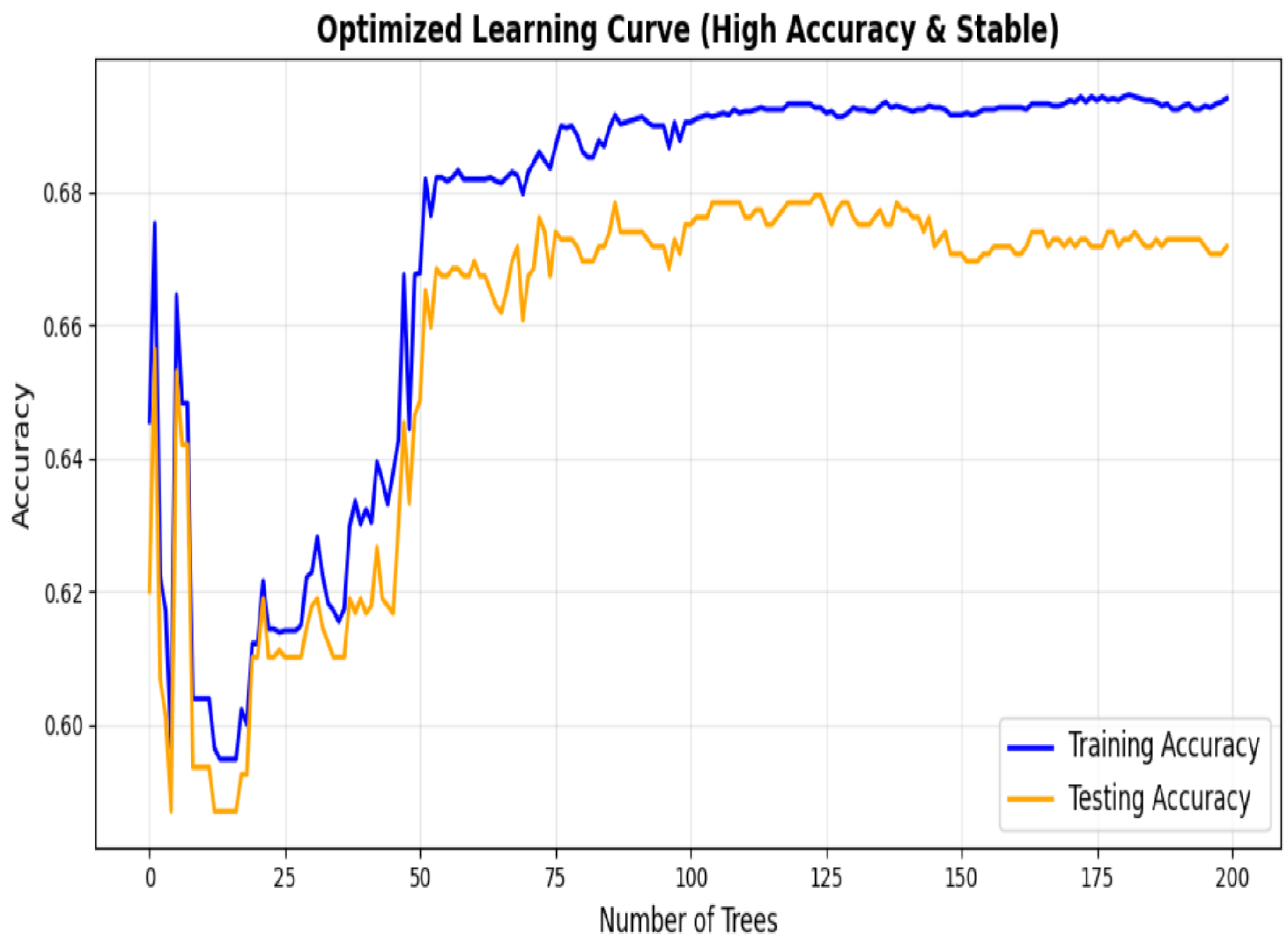
XGBOOST CLASSIFICATION MODEL

** **Feature Engineering:** Concentrating on 4 core pillars: Semantic Similarity, Total Experience, Required Experience, and the Experience Gap.

** **Binary Suitability Logic:** Transforming scores into a clear (Suitable vs. Not Suitable) decision using a 60% threshold.

** **Class Balancing Strategy:** Implementing dynamic weighting (scale_pos_weight) to ensure high sensitivity toward top-tier candidates despite their scarcity in the dataset.

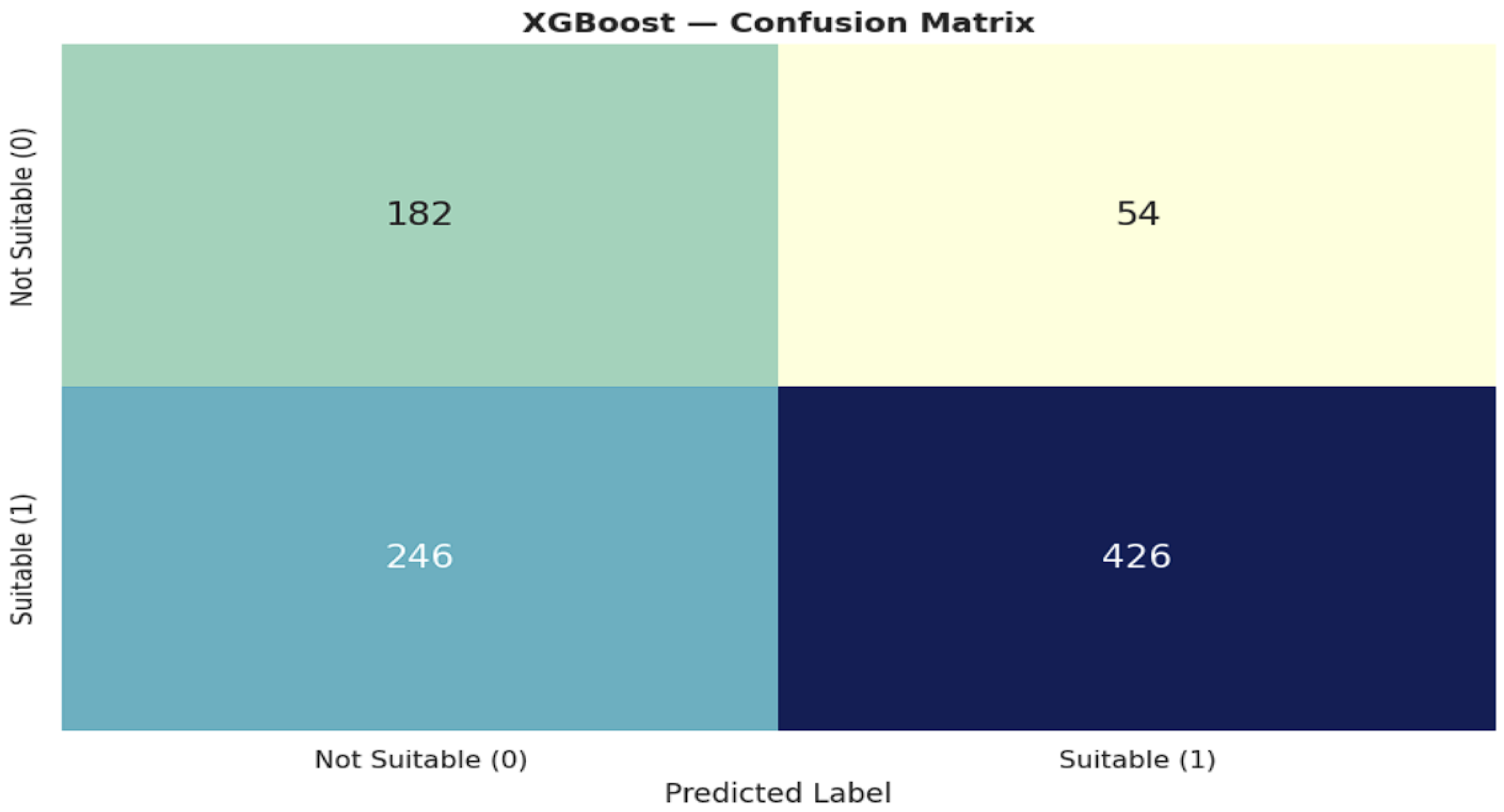
** **Robust Generalization:** Applying low learning rates and shallow tree depths to prevent overfitting, ensuring the model performs reliably on unseen resumes.



f1-score

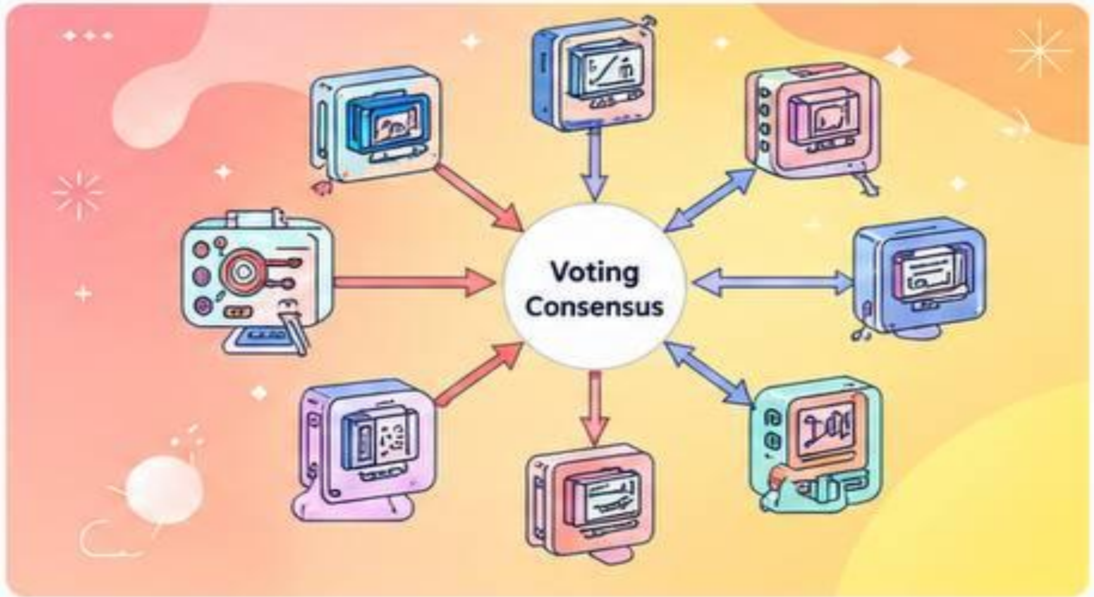
Suitable = 0.89%

Not Suitable = 0.43%


XGBoost — Confusion Matrix
Not Suitable (0)
Suitable (1)
182
54
246
426
Not Suitable (0)
Suitable (1)
Predicted Label

Voting Model (Ensemble Voting)

Voting is an **ensemble technique** that combines the predictions of **three or more base models** to improve overall performance.



Improved Accuracy

Combines multiple models to boost predictive performance and reduce overfitting.

Robust and Reliable

Reduces the impact of individual model errors through aggregated decisions.

Versatile

Can be applied across different algorithms and diverse datasets.

Random Forest

Reduces variance through bagging (bootstrap aggregating).

XGBoost

Optimized for efficiency and handles non-linear relationships effectively.

Gradient Boosting

Focuses on reducing bias by learning from previous errors sequentially.

Key Idea: By leveraging the strengths of different models, Voting achieves better generalization, stability, and accuracy.

X_smart

The 5 Smart Features powering the Voting Classifier

Why only 5 features?

Using all 384 SBERT dimensions causes Overfitting. These 5 capture the full signal without the noise.

01 semantic_similarity

Full semantic match between the CV and job description — computed via Cosine Similarity on SBERT vectors.

- Captures meaning beyond keywords

02 skills_jaccard

Candidate skills ∩ required skills ÷ their union (Jaccard Index). Directly measures hard-skill overlap.

- Most direct suitability signal

03 years_experience

Total real work experience — calculated from start/end dates across all the candidate's listed roles.

- Quantifies seniority objectively

04 required_years

Experience required by the job posting — auto-extracted via Regex from the description text.

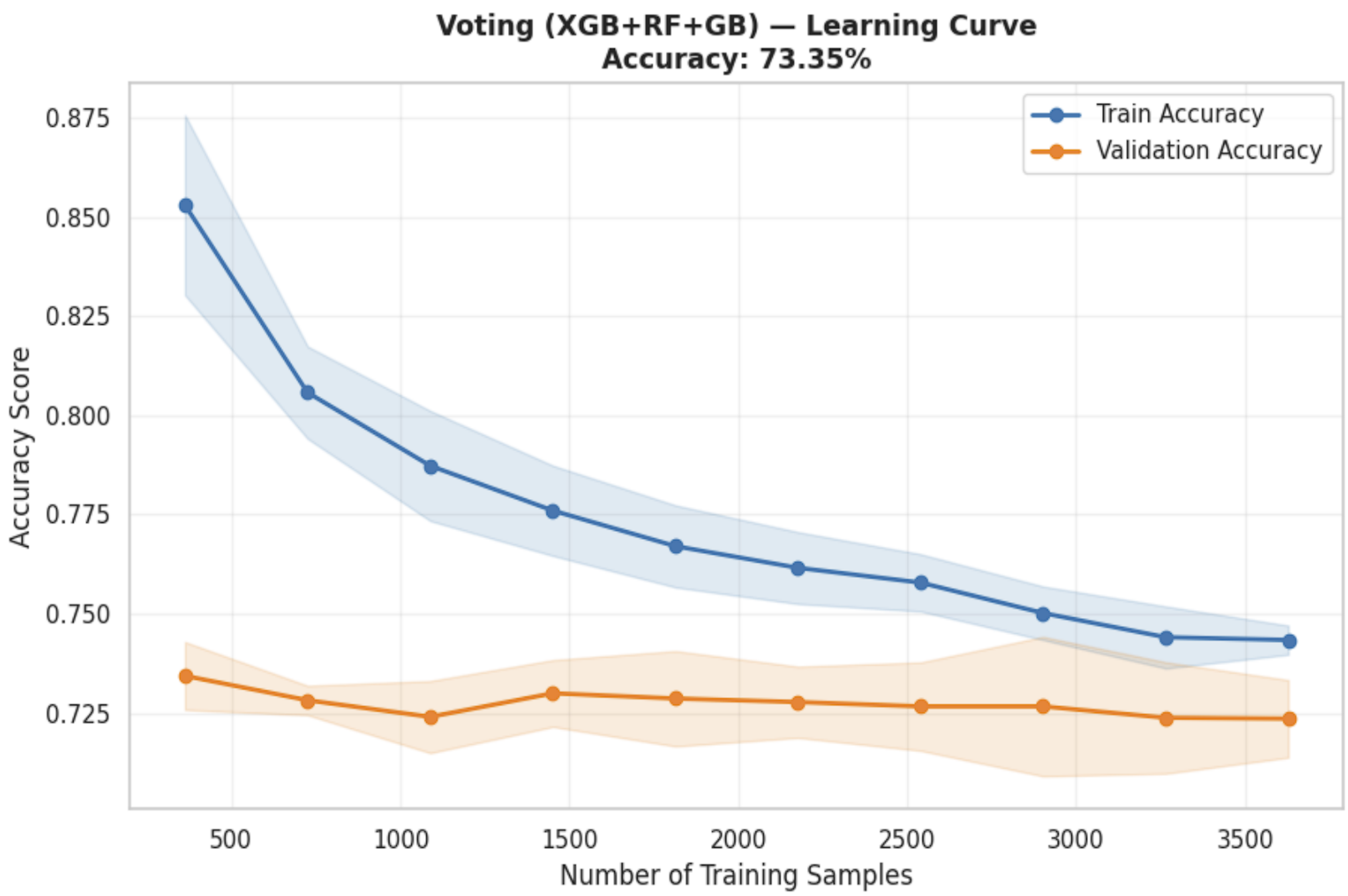
- Sets the benchmark to compare against

05 exp_gap

Difference: candidate's experience minus required years (can be negative). Reveals junior / matched / over-qualified.

- Critical for ranking at the right tier

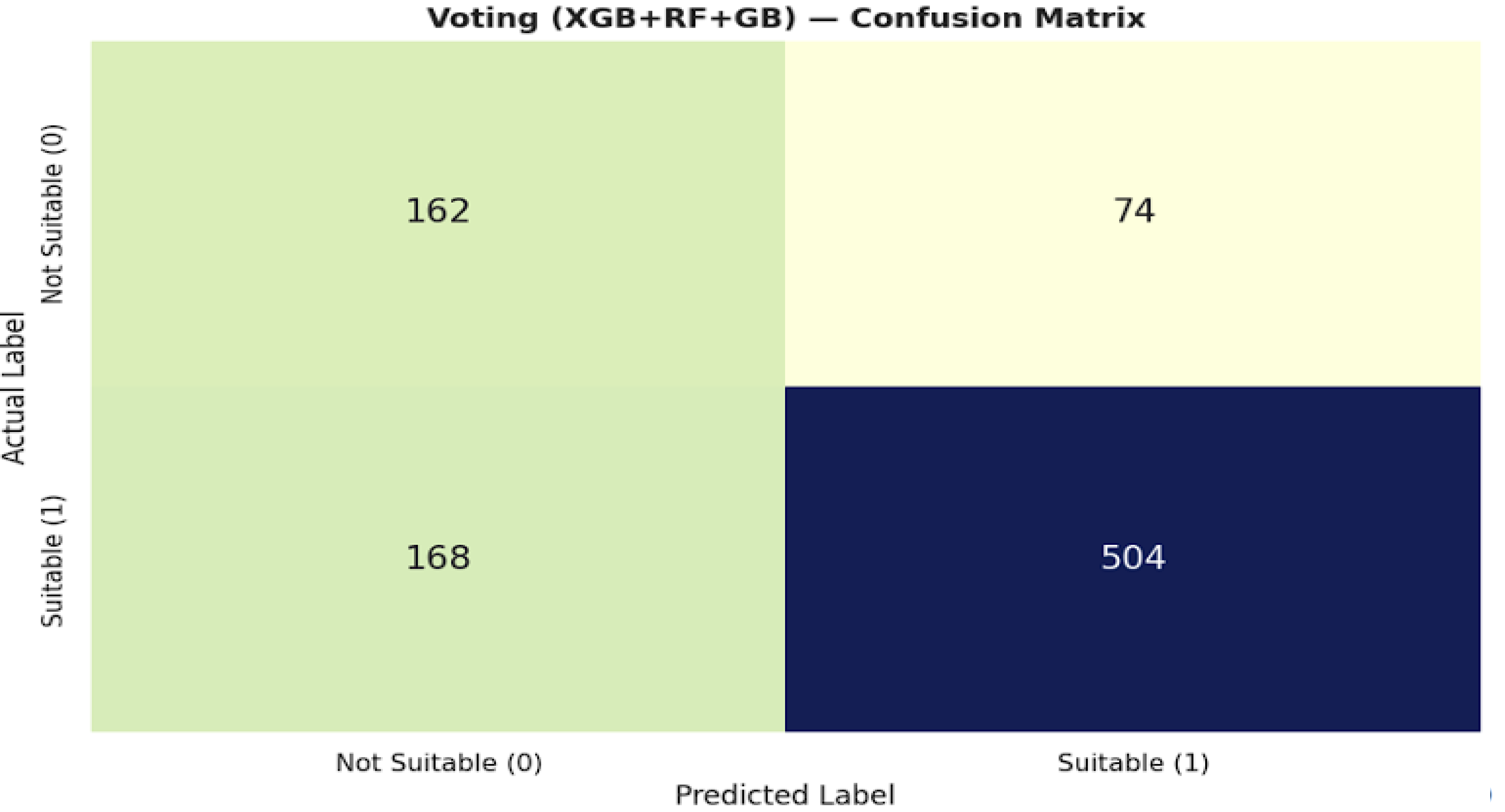
Key Idea: `X_smart = [ semantic_similarity | skills_jaccard | years_experience | required_years | exp_gap ]`



f1-score

Suitable = 0.87%

Not Suitable = 0.49%


Voting (XGB+RF+GB) — Confusion Matrix
Not Suitable (0)
Suitable (1)
Actual Label
162
74
168
504
Not Suitable (0)
Suitable (1)
Predicted Label

The reviewer

Resume Dataset by Saugata Roy Arghya (Kaggle)

Application of LLM Agents in Recruitment (2024)

Smart Resume Screening Tool (2023)

Enhanced Resume Screening for Smart Hiring (2024)

Resume Screening with NLP (2024)

A novel approach for detecting resume columns (2025)

Enhanced Resume-Job Matching via Graph-based Networks (2024)